

ANIMESH BANGAL

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PROFESSIONAL SUMMARY

Mechanical Engineering undergraduate at LSU with a 4.21 GPA and a Minor in Robotics, specializing in controls, UAV dynamics, autonomous systems, robotics, propulsion research, and data acquisition. Experienced in physics-based Simulink modeling, PID and Model Predictive Control (MPC), optimal control, input shaping, computer vision, experimental instrumentation, fuel characterization, and autonomous systems development across NASA-sponsored and NSF-funded research programs. Author on four controls research papers, including one published at the LSU Symposium on Control, Learning & Intelligent Systems and four submissions to the American Control Conference (ACC) 2027.

EDUCATION

Bachelor of Science, Mechanical Engineering, Louisiana State University (LSU)
Minor: Robotics
GPA: 4.21 —

Expected May 2028
Baton Rouge, LA

EXPERIENCE

Louisiana State University – Stein’s Lab (iCORE)
Undergraduate Researcher

Baton Rouge, LA
Nov 2025 – Present

- Conduct research in nonlinear control, input shaping, optimal control, UAV dynamics, trajectory tracking, and autonomous perception for aerial robotic systems.
- Contributed to the analytical derivation of closed-form nonzero-initial-condition input-shaping methods for vibration suppression and robust motion control.
- Led the experimental implementation of the *Optimal Periodic Control* project, progressing from dynamic modeling and simulation to controller development, hardware integration, and experimental validation on a Flightory Stallion fixed-wing UAV.
- Built and instrumented the Flightory Stallion experimental platform and integrated flight-control hardware, onboard electronics, and supporting systems for closed-loop control experiments.
- Developed physics-based Simulink and state-space UAV models and implemented PID, intelligent PID (iPID), and Model Predictive Control (MPC) algorithms for stabilization and trajectory tracking.
- Currently developing a computer-vision-based perception system for a quadcopter to detect and track a target UAV and estimate and record its trajectory for experimental interception and control research.

LaACES – NASA-Sponsored Program, Louisiana State University
LaACES Student Researcher

Baton Rouge, LA
Jul 2025 – May 2026

- Designed and integrated an experimental payload for measuring ultraviolet irradiance during a NASA-supported high-altitude balloon mission.
- Integrated a UV photodiode sensor and signal-conditioning electronics using a transimpedance amplifier for measurement.
- Developed Arduino-based flight software for sensor acquisition, data logging, and payload operation during the high-altitude balloon mission.
- Performed electrical testing, sensor calibration, and system-level validation to verify reliable payload operation prior to flight.
- Built MATLAB scripts to process flight data and analyze UV intensity trends with altitude for operation at approximately 30–35 km.

Louisiana State University – Jet & Propulsion Lab
Undergraduate Researcher

Baton Rouge, LA
Jun 2026 – Present

- Conduct experimental combustion and sustainable aviation fuel research involving flame behavior, fuel characterization, and emissions analysis.
- Perform combustion experiments using a McKenna burner to investigate flame stability and gaseous fuel behavior under controlled operating conditions.
- Analyze soot emissions from Sustainable Aviation Fuel (SAF) and Jet A using laser-based optical diagnostic techniques.
- Gain hands-on experience with comprehensive two-dimensional gas chromatography (GC×GC) for fuel characterization and compositional analysis.
- Work with analytical techniques including gas chromatography with flame ionization detection (GC-FID) and vacuum ultraviolet (VUV) spectroscopy for hydrocarbon identification and fuel characterization.
- Support experimental setup, instrumentation, sample preparation, data acquisition, and analysis for combustion and fuel-characterization studies.
- Research biomass-to-aviation-fuel pathways and refinery integration as part of an NSF-funded sustainable propulsion project.

Tiger Propulsion, Louisiana State University
Data Acquisition (DAQ) Lead

Baton Rouge, LA
2026 – Present

- Lead development and system integration of the data acquisition architecture for propulsion testing, coordinating sensors, signal conditioning, hardware interfaces, and data logging across multiple subsystems.
- Select and evaluate instrumentation including load cells, pressure transducers, temperature sensors, and other measurement hardware for monitoring propulsion-system performance.

- Develop sensor integration plans and measurement requirements to capture thrust, pressure, temperature, load, and transient response data during subsystem and propulsion testing.
- Process and analyze experimental data in MATLAB to generate engineering plots and performance curves including thrust-time, pressure-time, and transient load/impact responses.
- Coordinate with propulsion, structures, and ground-support teams to establish sensor placement, sampling requirements, test objectives, and system interfaces.

Gulf Scholar, Louisiana State University

Research Scholar

Baton Rouge, LA
May 2026 – Aug 2026

- Designed, built, and tested an autonomous quadcopter for automated water sample collection in Gulf wetlands.
- Developed and integrated a UAV-mounted sampling mechanism for autonomous water collection missions.
- Integrated sensors, actuators, Pixhawk flight-control hardware, and Raspberry Pi computing hardware into the autonomous platform.
- Programmed autonomous flight paths and coordinated flight-control and payload operations for repeatable sampling missions.
- Conducted system integration, troubleshooting, and field tests at LSU lakes and nearby wetlands.

PERSONAL PROJECTS

Open-Source 6-DOF Robotic Arm with Input Shaping and Computer Vision

Personal Project

- Building and programming an open-source 6-DOF robotic arm for autonomous pick-and-place manipulation.
- Developing kinematic and dynamic models to support trajectory planning, motion control, and simulation of the robotic manipulator.
- Implementing input-shaping techniques to reduce residual vibration during object transfer and achieve smoother end-effector motion.
- Developing a computer-vision perception pipeline to recognize target objects and determine which object the robotic arm should pick up.
- Integrating perception, trajectory generation, and motion control to autonomously move selected objects between locations with minimal residual vibration.

PUBLICATIONS

- **Drone Interception via EDMDC-Based Model Predictive Control** – Third Author. Published at the LSU Symposium on Control, Learning & Intelligent Systems; submitted to the American Control Conference (ACC) 2027.
- **Optimal Periodic Control** – Second Author. Submitted to the American Control Conference (ACC) 2027.
- **Adaptive Nonzero-Initial-Condition Input Shaping with Online Modal Identification** – Third Author. Submitted to the American Control Conference (ACC) 2027.
- **Closed-Form Development for Robust Nonzero-Initial-Condition Input Shaping** – Third Author. Submitted to the American Control Conference (ACC) 2027.

ACTIVITIES & LEADERSHIP

- **American Society of Mechanical Engineers (ASME), LSU Chapter**
- Student Volunteer, American Control Conference (ACC) 2026, New Orleans, LA
- Volunteer and Reviewer, LSU Symposium on Control, Learning & Intelligent Systems, Baton Rouge, LA
- Volunteer, Summer Undergraduate Research Forum (SURF), LSU

AWARDS & HONORS

- President's List: Fall 2024, Spring 2025
- Dean's List: Spring 2024, Fall 2025, Spring 2026
- BASF Team Chemistry Award

SKILLS

Controls & Simulation: MATLAB, Simulink, PID Control, Intelligent PID (iPID), Model Predictive Control (MPC), State-Space Modeling, Optimal Control, Input Shaping, Trajectory Tracking, Autonomous Systems

Programming & Embedded Systems: Python, Arduino, Raspberry Pi, Pixhawk, Computer Vision

Data Acquisition & Instrumentation: DAQ System Integration, Load Cells, Pressure Transducers, Sensor Integration, Signal Conditioning, Experimental Data Analysis

Experimental & Fuel Analysis: GC×GC, GC-FID, VUV Spectroscopy, Combustion Testing, Laser-Based Optical Diagnostics, Fuel Characterization

CAD & Design: AutoCAD (Certified), SolidWorks (Certified), Fusion 360

Optimal Periodic Control

Second Author

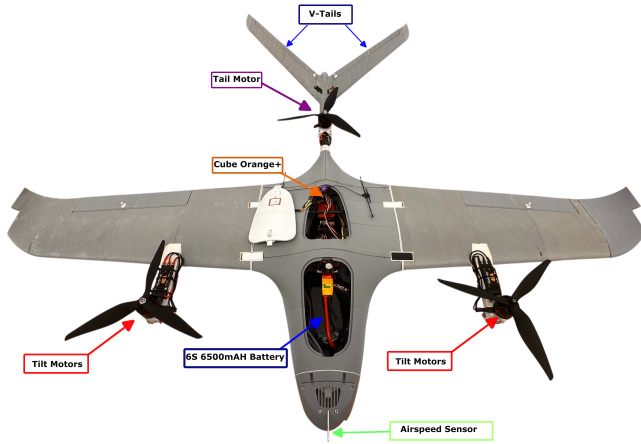


Figure 1: Flightory Stallion hardware. Laboratory-built tiltrotor VTOL platform with labeled propulsion, flight controller, battery, and airspeed sensor. Used for preliminary VTOL tests; optimized wing-borne periodic flight remains future work. Source: fixed-wing manuscript, Fig. 6.

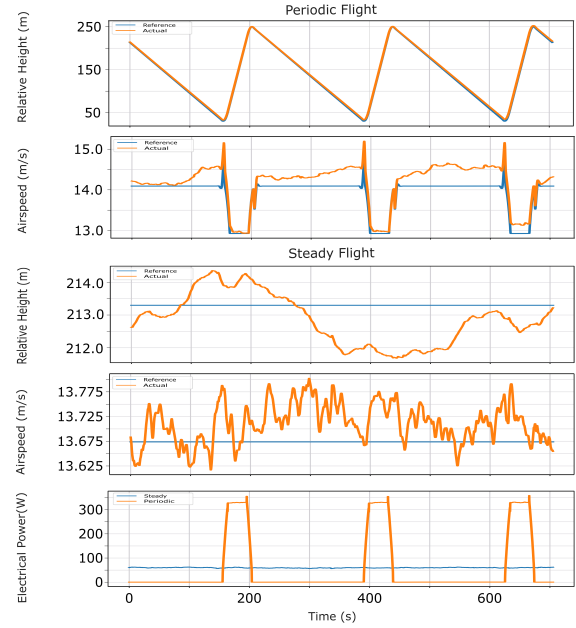


Figure 2: Optimal control simulation. Gazebo altitude and airspeed tracking for periodic and steady flight, with electrical propulsion power estimated from thrust and airspeed. The paper reports a 2.182% simulated power saving under constant efficiency. Source: fixed-wing manuscript, Fig. 5.

Drone Interception via EDMDc-Based Model Predictive Control

Third Author

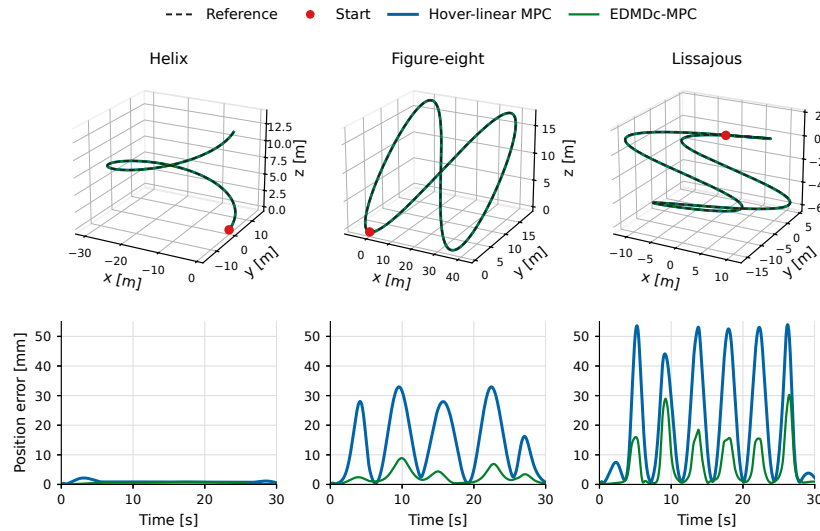


Figure 3: Matched trajectory-tracking simulations. Reference paths, hover-linear MPC, and EDMDc-MPC are compared for helix, figure-eight, and Lissajous trajectories; lower panels show position-error histories. Under matched tracking settings, the paper reports mean coordinate position RMSEs of 6.958 mm for hover-linear MPC and 1.576 mm for EDMDc-MPC across 200 paired scenarios. Source: drone-interception manuscript, Fig. 2.